Integration: Lengths

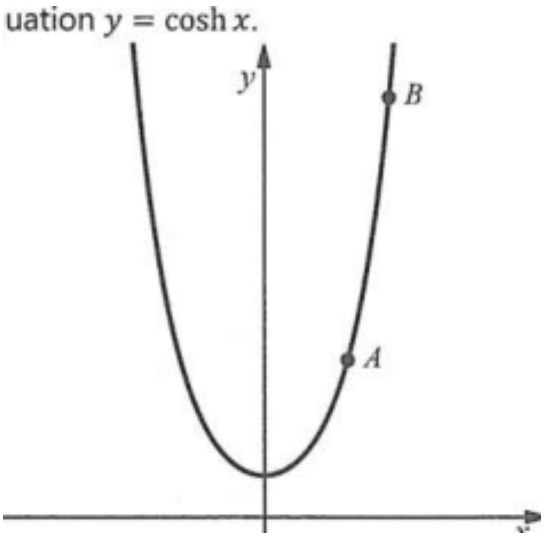
Questions 1–25

Question 1. Two points P and Q lie on the curve given. In each case the x-coordinate of P and Q are given along with the curve. Find the exact length of the arc PQ.

a) $x_P = 8$, $x_Q = 35$, $y = \frac{2}{3}x^{3/2}$

b) $x_P = 1$, $x_Q = 4$, $y = \frac{4}{3}x^{3/2}$

Question 2. The diagram below shows the curve with equation $y = \cosh x$.



The points A and B have x-coordinates 2 and 3 respectively. Show that the arc length of the curve from A to B is equal to $\frac{7}{12}$.

Question 3.

a) Show that $\int \sec x \, dx = \ln|\sec x + \tan x|$.

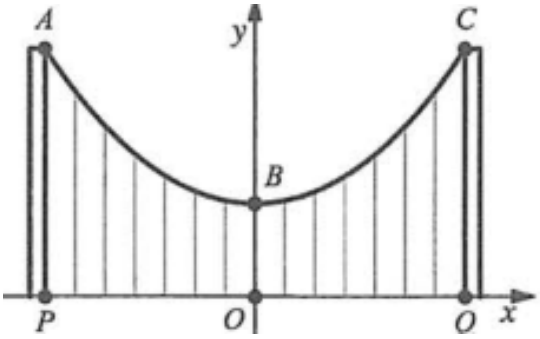
A curve is given for $\frac{\pi}{4} \leq x \leq \frac{\pi}{3}$ by the equation $y = \ln(\cos x)$.

b) Find the length of the arc on this curve, s, giving your answer in the form $\ln\left(\frac{p}{q}\right)$ where p and q are exact irrational numbers.

Question 4. A curve is given by the equation $y = \frac{1}{2}x^2$. Find the exact length of the arc on the curve, s, between the origin and the point where $x = 2$.

Question 5. A curve is given by the equation $x = 2\cosh\left(\frac{y}{2}\right)$. Find the exact length of the arc on the curve, s, between the points where $y = 0$ and $y = \ln 4$.

Question 6. Part of a suspension bridge is shown in the diagram below, where P and Q are directly beneath A and C respectively.



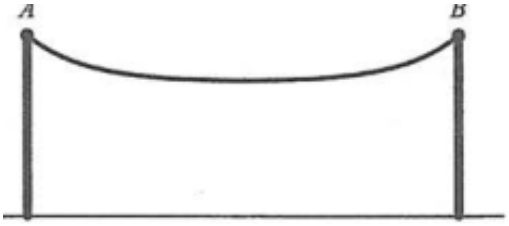
It is given that $PO = OQ = 70$ metres, and $AP = CQ = 40$ metres. Point B is 15 metres above O. The cable that runs from A to B to C can be modelled by the equation $y = k\cosh\frac{x}{k} - p$, where k and p are constants.

a) Show that $k - p = 15$.
b) Use part (a) to verify that $k = 101.91$ to two decimal places.
c) Find the length of the cable, from A to C, to the nearest metre.

Question 7. Two masts, A and B, are built 500 metres apart on horizontal ground. A wire is hung between the masts so that its shape can be modelled by the equation

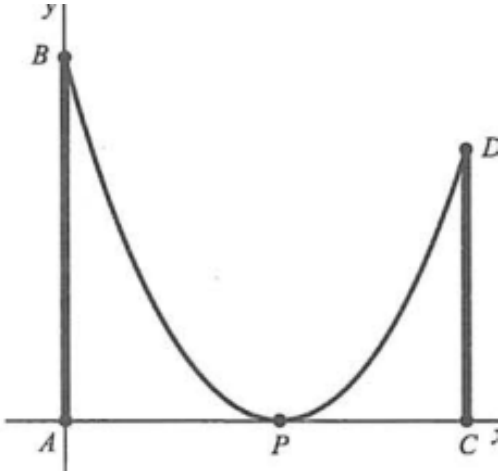
$$y = p + \cosh\left(\frac{x}{q} - r\right)$$

where y is the height in metres of the wire above the ground, x is the distance in metres from mast A, and p, q and r are positive constants. The diagram below shows the model as being hung from the very top of the mast, which is at a height of 80 metres, and that the minimum distance between the ground and the wire is 60 metres.



a) Find the value of p, and hence find q and r to three significant figures.
b) Find the length of the wire hung between A and B, giving your answer to the nearest metre.

Question 8. Two poles AB and CD are placed in the ground, 3 metres apart from each other, as shown in the diagram below. The height of the pole AB is 1.5 metres.



A rope is attached to B and D, the tops of the two poles, and hangs so that it touches the ground at point P, which is 1.6 metres from A. The curve is modelled using the equation

$$y = p\cosh(qx - r) - 2$$

where p, q and r are positive constants.
a) Find the value of p, and hence find q and r to four significant figures.
b) Find the height of pole CD, giving your answer to the nearest centimetre.
c) Find the length of the rope between B and D, giving your answer to the nearest centimetre.

Question 9. For each of the following, find the length of the arc, s, on the given curve between the origin and the point at $x = 1$, giving your answer in terms of the constant, $k > 0$:

a) $y = \frac{1}{k}\cosh^2(kx)$

b) $y = \frac{1}{k}\sinh^2(kx)$

Question 10.

a) By using the substitution $x = \frac{1}{6}\sinh u$, show that

$$\int \sqrt{1+36x^2}\,dx = \frac{1}{12}\operatorname{arsinh}(6x) + \frac{1}{2}x\sqrt{1+36x^2} + c.$$

Hence calculate the exact length of the arc on the curve $y = 3x^2$ between the points $x = 0$ and $x = \frac{1}{3}$.

Question 11.

a) By using substitution $x = \sinh^2 u$ show that

$$\int \sqrt{\frac{x+1}{x}}\, dx = \sqrt{x(x+1)} + \operatorname{arsinh} \sqrt{x} + c.$$

Hence calculate the exact length of the arc, s, on the curve $y = \frac{2}{3}\sqrt{x}$ between the points $x = 1$ and $x = 4$, giving your answer in logarithmic form.

Question 12. A curve is given by the equation $y = 5 - \ln(1 - x^2)$ for $x \geq 0$.

a) Show that the exact length of the arc on the curve y is given by

$$s = \int \left(\frac{1+x^2}{1-x^2}\right) dx.$$

b) Hence the exact length of the arc on the curve between the points where $x = 0$ and $x = \frac{1}{2}$.

Question 13. A landscape gardener is designing a curved path which will be exactly 39 metres long. The edge of the path follows the curve $y = \frac{1}{3}x^{3/2}$ where x and y are measured in metres. By considering the origin as the start point of the curve, find the value of x which gives a length of 39 metres.

Question 14. Engineers are designing the shape of a suspension bridge cable. The cable hangs between two supports and its height above the road can be modelled by the equation

$$h = a + \cosh\left(\frac{x}{b} - c\right)$$

where h is the height of the cable above the road (in metres), x is the horizontal distance from one end of the bridge (in metres), and a, b and c are constants. The bridge span is 60 m long (horizontally), so $0 \leq x \leq 60$. The lowest point of the cable occurs at $x = 30$, where the height is 12 metres. The cable height at each end of the bridge is 20 metres.

a) Find the values of a, b and c giving your answer in exact form.

b) Hence, using your calculator, find the length of the cable used giving your answer to the nearest metre.

Question 15. A curve for $t \geq 0$ has parametric equations

$$x = t^2, \quad y = \frac{1}{3}t^3 - t.$$

a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (t^2+1)^2$.

b) Hence find the exact length of the curve between the points that occur when $t = 0$ and $t = 4$.

Question 16. A curve is defined by the parametric equations

$$x = \frac{2}{3}t^3 + 2, \quad y = 2\sqrt{2}\,t^2 + 1$$

for $0 \leq t \leq 2$.

a) Show that the arc length between $t = 0$ and $t = 2$ is given by $s = \int_0^2 2t\sqrt{t^2+k}\,dt$ for a constant value k.

b) Hence find the exact length of the arc of the curve between the points where $t = 0$ and $t = 2$.

Question 17. A curve is defined by the parametric equations

$$x = 5(1+t)^{1/2}, \quad y = 5(1-t)^{1/2}$$

for $0 \leq t \leq 1$. Find the exact length of the arc of the curve between the points $(5, 5)$ and $(5\sqrt{2}, 0)$.

Question 18. A circle has parametric equations

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta.$$

By finding the arc length of the curve from $\theta = 0$ to $\theta = 2\pi$, show that the length of the circumference is $2\pi r$.

Question 19. A curve is defined by the parametric equations

$$x = 2 + e^{-t}(\cos t + \sin t), \quad y = 1 + e^{-t}(\cos t - \sin t)$$

for $t \geq 0$. Find the exact length of the arc of the curve between the points where $t = 0$ and $t = 3$.

Question 20. A curve is defined by the parametric equations

$$x = 2 + 3\cos t, \quad y = 3t + 3\sin t$$

for $t \geq 0$.

a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 36\cos^2\left(\frac{t}{2}\right)$.

b) Hence find the exact length of the curve between the points that occur when $t = 0$ and $t = \pi$.

Question 21. A curve is defined by the parametric equations

$$x = 3\sin^3 t, \quad y = 3\cos^3 t$$

for $0 \leq t < 2\pi$. Find the exact length of the arc of the curve between the points where $t = 0$ and $t = \frac{\pi}{2}$.

Question 22. A curve is defined by the parametric equations

$$x = 4\sin t - \sin 4t, \quad y = 4\cos t - \cos 4t$$

for $t \geq 0$.

a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 64\sin^2\left(\frac{3t}{2}\right)$.

b) Hence find the exact length of the curve between the points that occur when $t = 0$ and $t = \frac{\pi}{3}$.

Question 23. A designer models the shape of a curved skateboard ramp using parametric equations

$$x = k(t + \sin t), \quad y = k(1 - \cos t)$$

where x and y are measured in metres, $t \geq 0$ and k is a positive constant. The ramp section runs from the ground at $t = 0$ to the platform at $t = \pi$. Given the exact length of the ramp between the ground and the ramp is 12 metres, find the value of k. **Prerequisite Knowledge for Question 24: Calculus of Hyperbolic Functions**

Question 24.

a) By using the substitution $u = \sinh x$, show that $\displaystyle\int \operatorname{sech} x \, dx = \arctan(\sinh x) + c$.

A curve is defined by the parametric equations

$$x = \operatorname{sech} t, \quad y = \tanh t$$

for $t \geq 0$.

b) Find the exact length of the arc of the curve between the points where $t = 0$ and $t = \ln 2$.

Question 25.

a) Show that

$$\int \sqrt{1+x^2}\,dx = \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln\left(x+\sqrt{x^2+1}\right) + c.$$

A curve is defined by the parametric equations

$$x = at^2, \quad y = 4at$$

for $t \geq 0$ where a is a positive constant.

b) Find the exact length of the arc of the curve between the points where $t = 0$ and $t = 1$ giving your answer in terms of a.